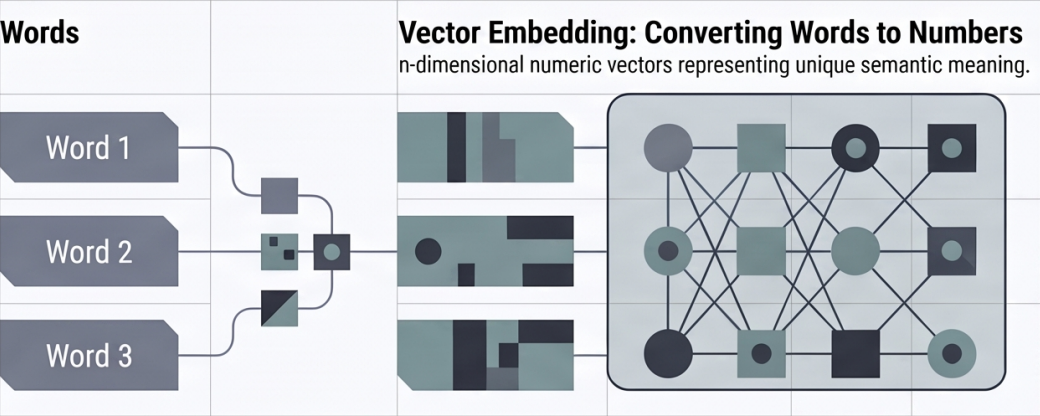
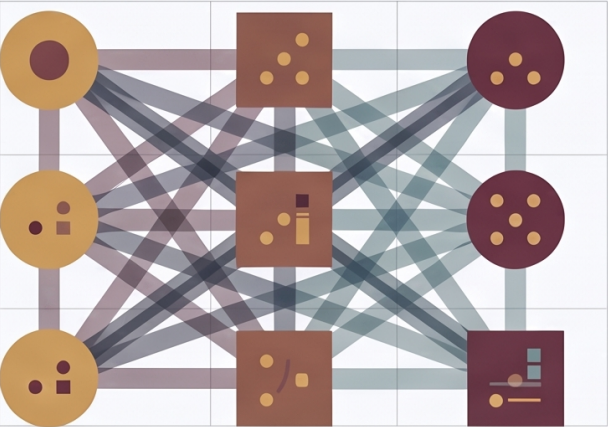


Inside the Transformer: The Three Pillars of LLM Intelligence

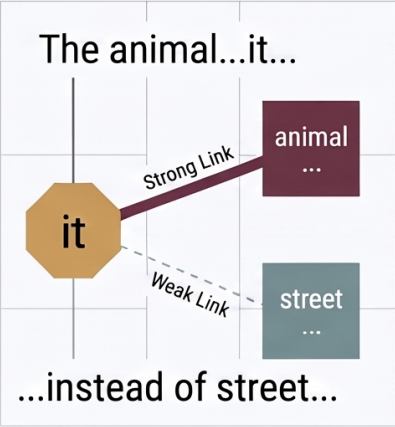
MAPPING & LINKING



Self-Attention: Connecting the Context
Every token looks at every other token simultaneously to dynamically link context-relevant words.

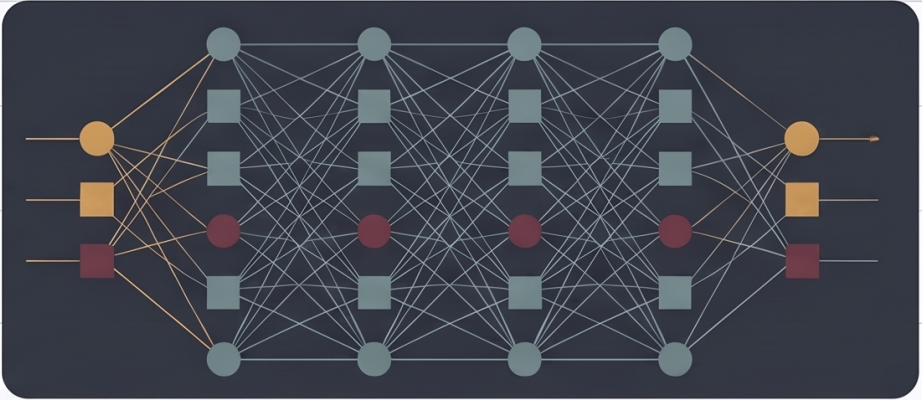


Resolving Ambiguity
Attention ensures the word 'it' links to 'animal' instead of 'street' in a sentence.



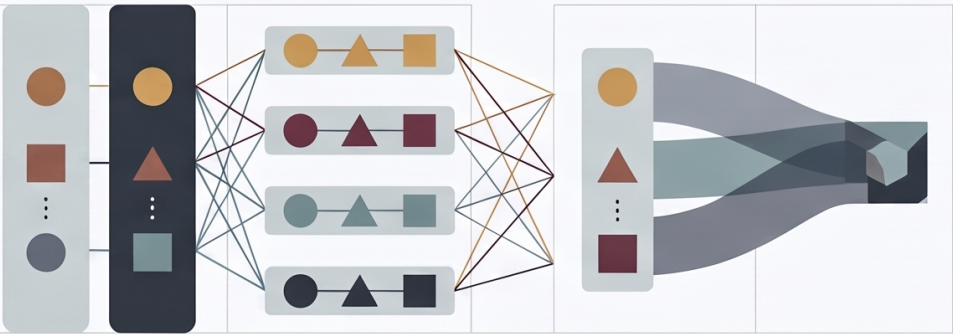
PROCESSING & STABILITY

FFN: The Model's 'Thinking' Space
Feed-Forward Networks store factual knowledge and account for roughly two-thirds of a model's memory.



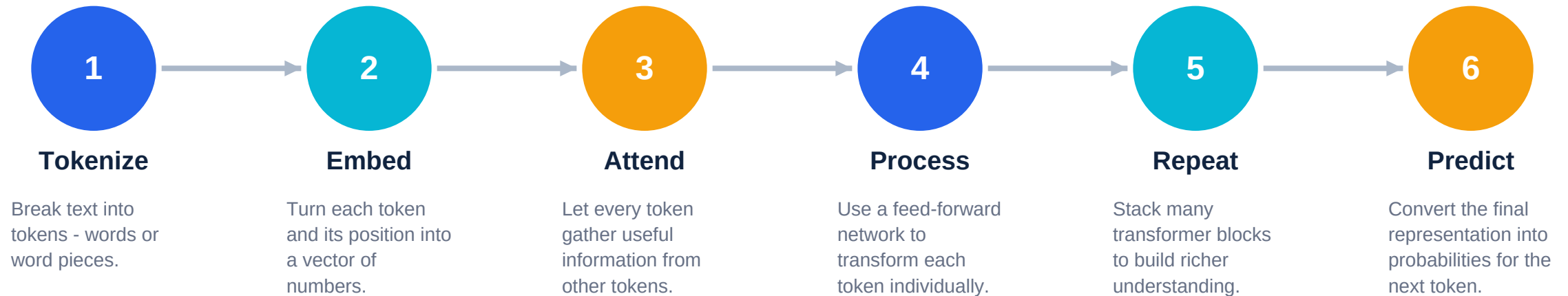
Maintaining Data Distinction
The FFN prevents token embeddings from collapsing, preserving individual word information.

Consolidation and Flow
After attention links words, the FFN processes each position to generate a refined representation.



How a transformer works - the big picture

A transformer repeatedly turns token representations into more useful, context-aware representations, then predicts what should come next.



Simple example: "The animal crossed the street because it was tired."

Embedding gives every token a numeric starting point. Self-attention helps "it" connect strongly to "animal". The feed-forward network then refines the meaning at each token position. Repeating this process helps the model predict or generate the next token.

(a) Vector embedding

A vector embedding converts a token into a list of numbers that a neural network can calculate with.

1 What it does

A computer cannot directly process meaning from written words. An embedding gives each token a learned numeric representation, such as [0.12, -0.44, 0.81, ...].

2 What it captures

During training, tokens used in similar situations develop related representations. "Dog" and "puppy" may be closer than "dog" and "spreadsheet".

3 Position matters

Transformers also add position information. This helps distinguish "dog bites man" from "man bites dog", even though both contain the same words.

Beginner analogy: a meaning map

dog

puppy

cat

spreadsheet

invoice

Nearby points suggest related usage or meaning. The overall numeric pattern carries the information.

(b) Self-attention mechanism

Self-attention lets each token decide which other tokens are most useful for understanding its current context.

1 Query

What information is this token looking for? Think: the question the current token asks.

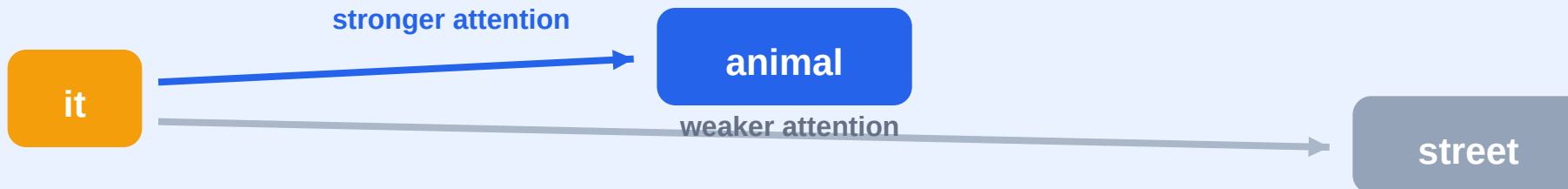
2 Key

What information does another token advertise? Think: a label describing what it can offer.

3 Value

What information should be passed along if the match is useful? Think: the actual content retrieved.

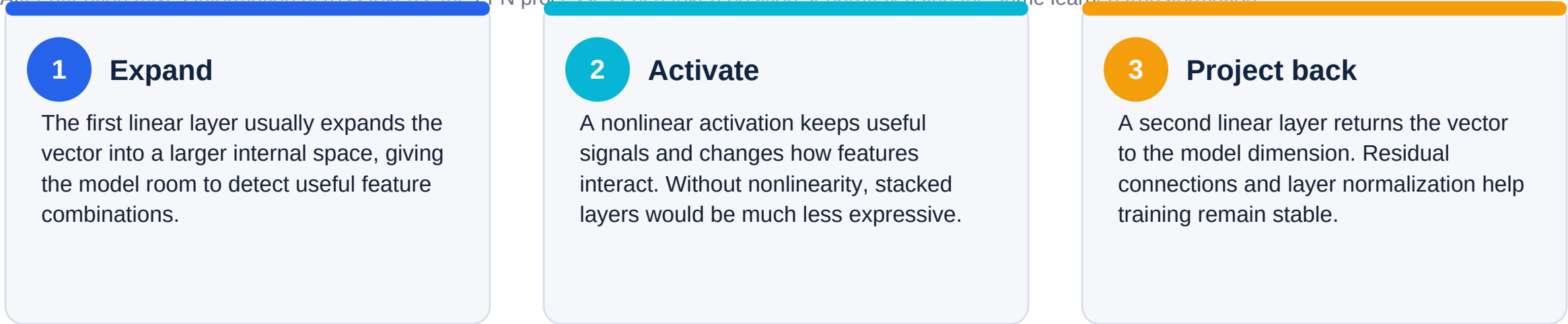
Example: "The animal crossed the street because it was tired."



The model calculates attention scores, turns them into weights, and combines the value vectors. Multi-head attention repeats this with several learned perspectives, so one head may track references while another tracks syntax or position.

(c) Feed-forward neural network

After attention mixes information across tokens, the FFN processes each token position separately using the same learned transformation.



How the three parts work together

